Project(Phase-II)
Branch: CSE-Artificial Intelligence and Machine Learning
Department: Data Engineering

Semester: VIII **Regulation: A3** **Code: A3CIP604**

Project Title: Sound-Aware Emergency Response Bot With Intelligent Alerting System(Safety Bot)

Batch Number: 04

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Abstract

Emergency situations often require immediate assistance, yet individuals may be unable to seek help due to panic, physical limitations, or sudden health conditions. Existing emergency systems largely depend on manual user interaction, making them ineffective during critical moments. The presence of false alarms and misuse further reduces system reliability. This project focuses on analyzing the limitations of current emergency response approaches. It highlights the need for an intelligent voice-based emergency monitoring system. Automated emergency detection can significantly reduce response time. Intelligent voice-driven technologies can improve emergency preparedness and response effectiveness.

Introduction

Emergency situations such as accidents, medical emergencies, or personal safety threats often demand immediate assistance. However, during critical moments, individuals may be unable to seek help due to panic, physical injury, or sudden health issues. Most existing emergency response systems rely on manual user intervention, which limits their effectiveness in such situations. Additionally, the lack of intelligent voice-based detection and proper safeguards against false alarms or misuse further reduces system reliability. These challenges highlight the need for a more autonomous and dependable emergency monitoring approach that can enhance timely response and improve overall safety.

Problem Statement

In emergencies, people may be unable to call for help due to panic or physical limits. Existing systems lack intelligent voice-driven detection, classification, and alerting, and are weakened by no safeguards against false alarms or misuse.

Project Objectives

- To provide intelligent, voice-driven assistance during emergency situations.
- To detect distress keywords and classify situation severity using machine learning.
- To identify emergency sounds such as sirens, glass breaking, and gunshots through audio analysis.
- To automatically alert family members or emergency services during critical conditions.
- To reduce false alarms by implementing misuse-prevention and safe testing mechanisms.

Project Scope

- The project focuses on an AI-enabled, voice-driven emergency monitoring system.
- Utilizes speech recognition and audio processing techniques to detect distress signals and emergency sounds.
- Applies machine learning models to classify emergency severity levels.
- Supports automated alerting mechanisms to notify predefined contacts or emergency services.
- Includes false alarm prevention and testing modes to ensure system reliability.

Literature Review/ Background Study

Research Paper Title	Scope / Key Features	Limitations in Context of Safety Bot
Emergency Sound Classification and Visual Alert System for Enhanced Situational Awareness Riza Kamelia, Hendra Kusuma (2025)	Emergency sound recognition with on-screen visual alerts	Lacks automated alert notification (SMS/contacts) for immediate response
Deep Neural Network-Based Indoor Emergency Awareness Using Contextual Information From Sound, Human Activity, and Indoor Position Hyoung-Gook Kim, Gee Yeun Kim (2020)	Multimodal emergency detection using sound, activity recognition, and indoor localization	High system complexity and infrastructure dependency, limiting real-time personal deployment
AI-Enhanced Emergency Call Centre Assistant and Situational Awareness for Optimized Emergency Response Annmaria Mathew Shibu et al. (2025)	Speech recognition and NLP-based assistance for emergency call handling	Relies on active user interaction; ineffective in unattended or unconscious scenarios
Self-Attentive Residual Temporal Convolutional Network-Based Indoor Emergency Awareness Using Contextual Information From Sound and Human Activity Ju-Hwan Lee, Hyoung-Gook Kim (2024)	Deep learning-based sound and activity analysis for indoor emergency awareness	Computationally intensive and unsuitable for lightweight, low-resource safety systems

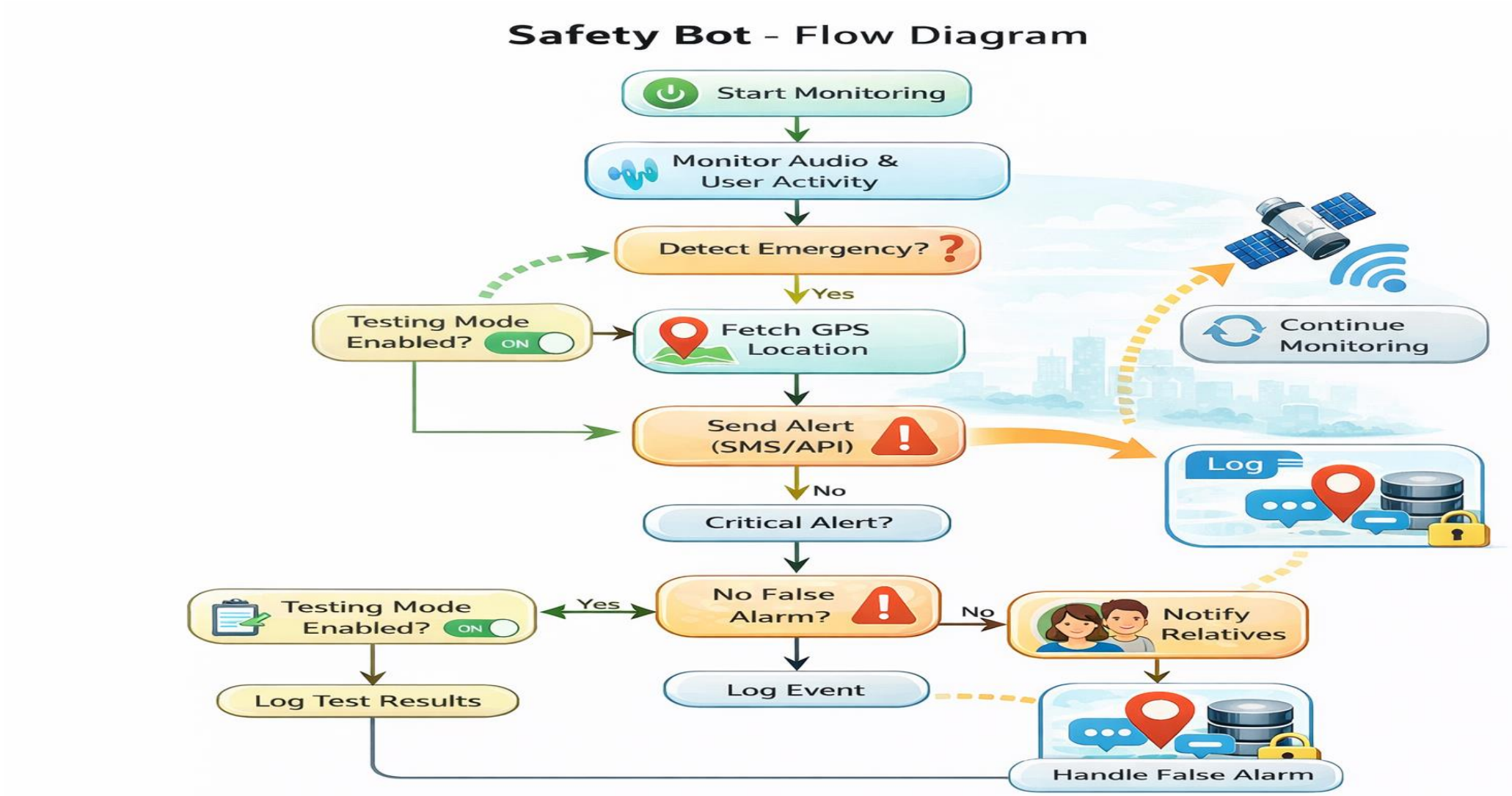
Data Gathered/ Data Used

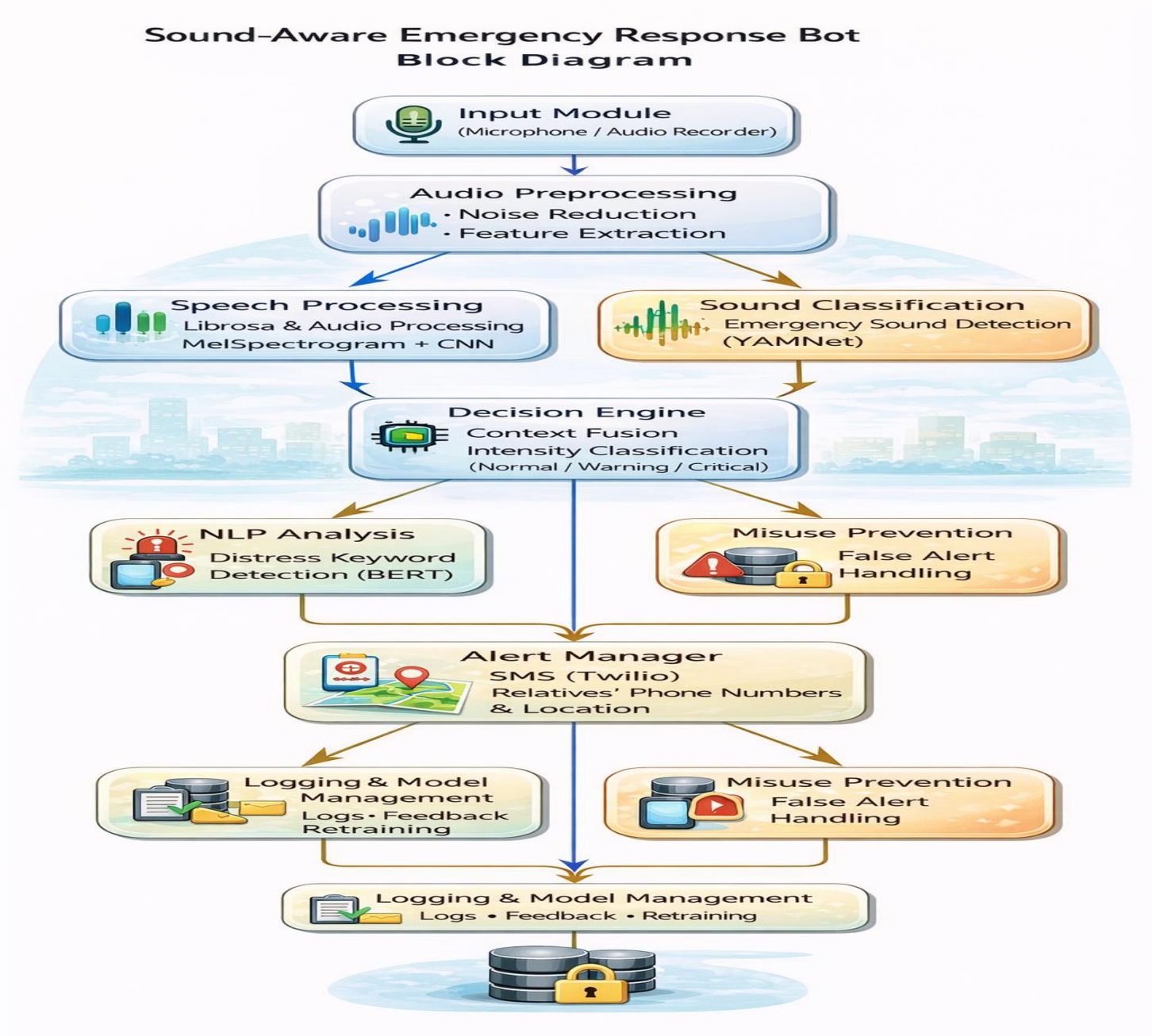
FSD50K:

FSD50K is a large-scale, open-source audio event classification dataset developed to support research in sound recognition and acoustic event detection. It contains diverse real-world audio recordings covering a wide range of sound categories, making it suitable for detecting emergency-related and distress sounds.

The dataset consists of two main subsets: the Development (Dev) dataset and the Evaluation (Eval) dataset.

Methodology / System Design




Sound-Aware Emergency Response Bot
Block Diagram
Input Module
(Microphone / Audio Recorder)
Audio Preprocessing
· Noise Reduction
· Feature Extraction
Speech Processing
Librosa & Audio Processing
MelSpectrogram + CNN
Sound Classification
Emergency Sound Detection
(YAMNet)
Decision Engine
Context Fusion
Intensity Classification
(Normal / Warning / Critical)
NLP Analysis
Distress Keyword
Detection (BERT)
Misuse Prevention
False Alert
Handling
Alert Manager
SMS (Twilio)
Relatives' Phone Numbers
& Location
Logging & Model
Management
Logs • Feedback
Retraining
Misuse Prevention
False Alert
Handling
Logging & Model Management
Logs • Feedback • Retraining

Implementation / Modules

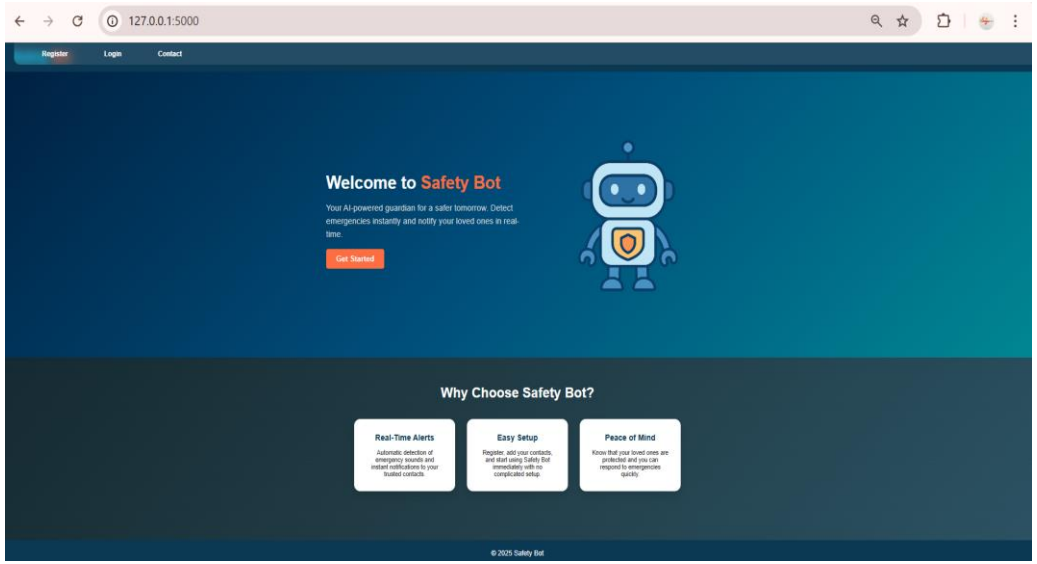
Link to code: https://github.com/satya-akkapu/Safety-Bot

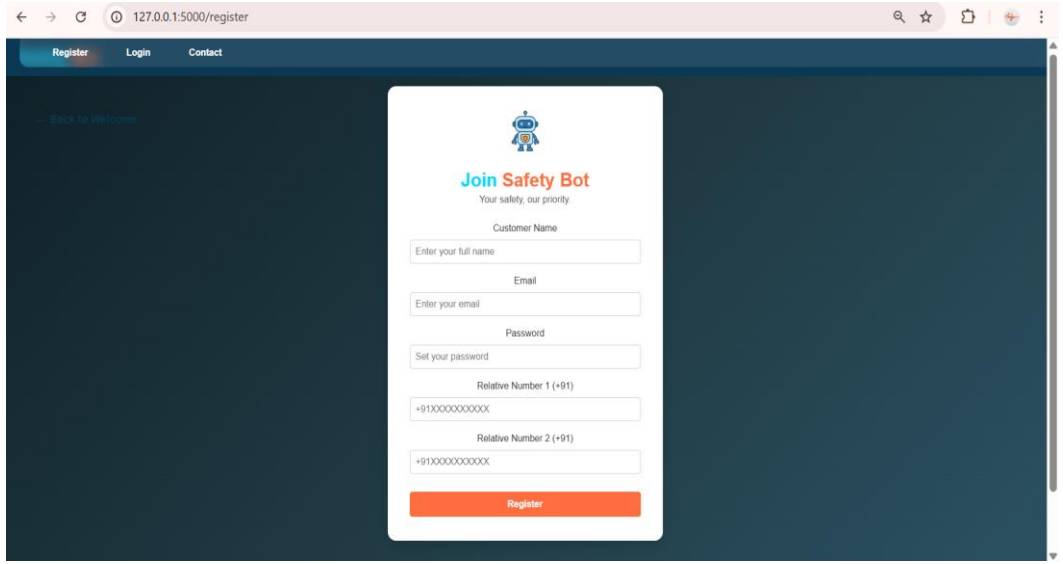
Key snippet:

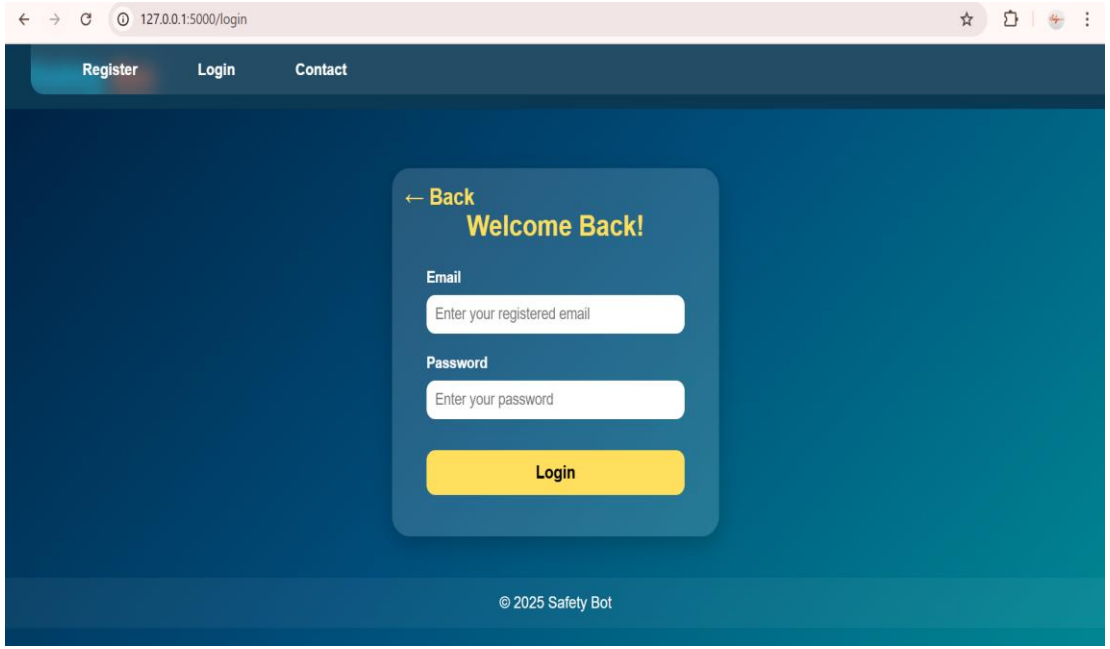
```
from flask import Flask, request, session, jsonify
from send_sms import send_emergency_sms
import sqlite3, os, uuid
app = Flask(__name__); app.secret_key="secret"
@app.route("/process_audio", methods=["POST"])
def process_audio():
    f = request.files["audio_data"]; path = os.path.join("audio_files", f"{uuid.uuid4()}.webm"); f.save(path)
    if True:  # demo emergency detection
        user = ...  # fetch user from DB
        send_emergency_sms(user[4], user[5])
        return jsonify({"status":"emergency_detected"})
    return jsonify({"status":"safe"})
```

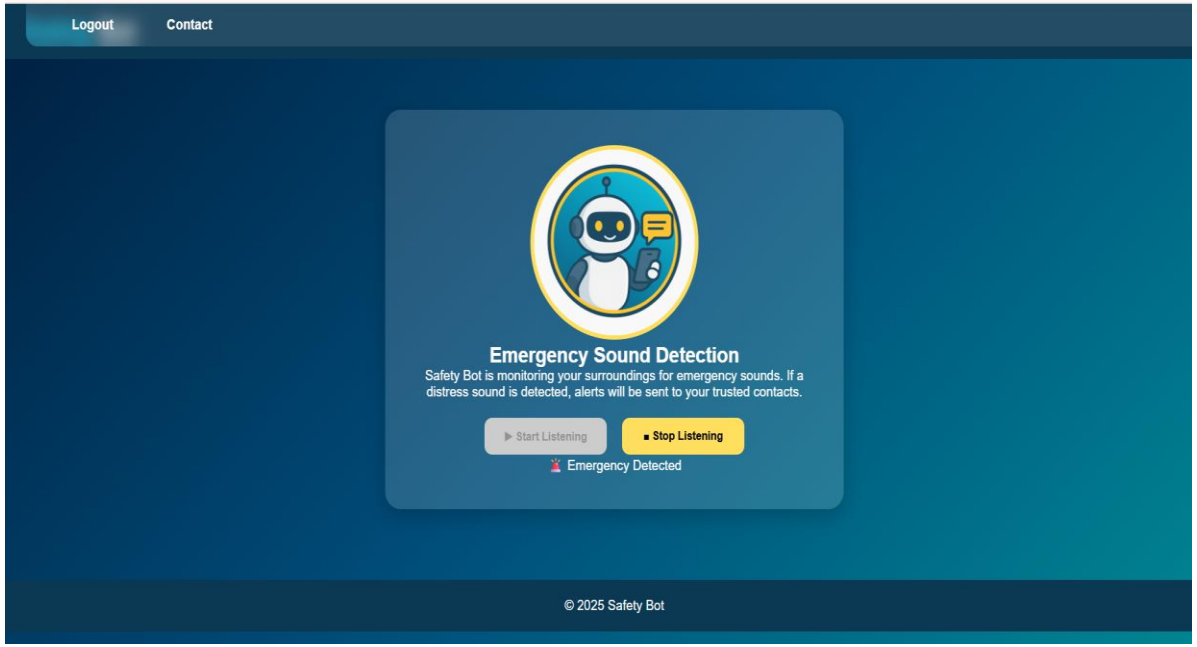
Results (Full) / Screenshots

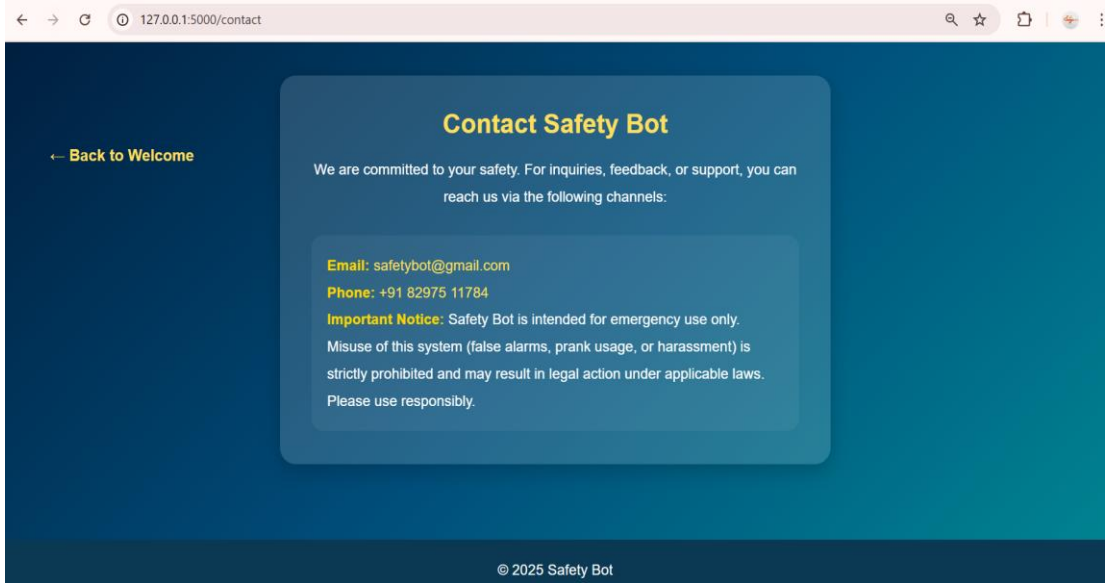
```
C:\Users\DELL\Downloads\Safety_bot>python app.py
✅ Safety Bot running at http://127.0.0.1:5000/
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
✅ Safety Bot running at http://127.0.0.1:5000/
 * Debugger is active!
 * Debugger PIN: 101-042-241
```



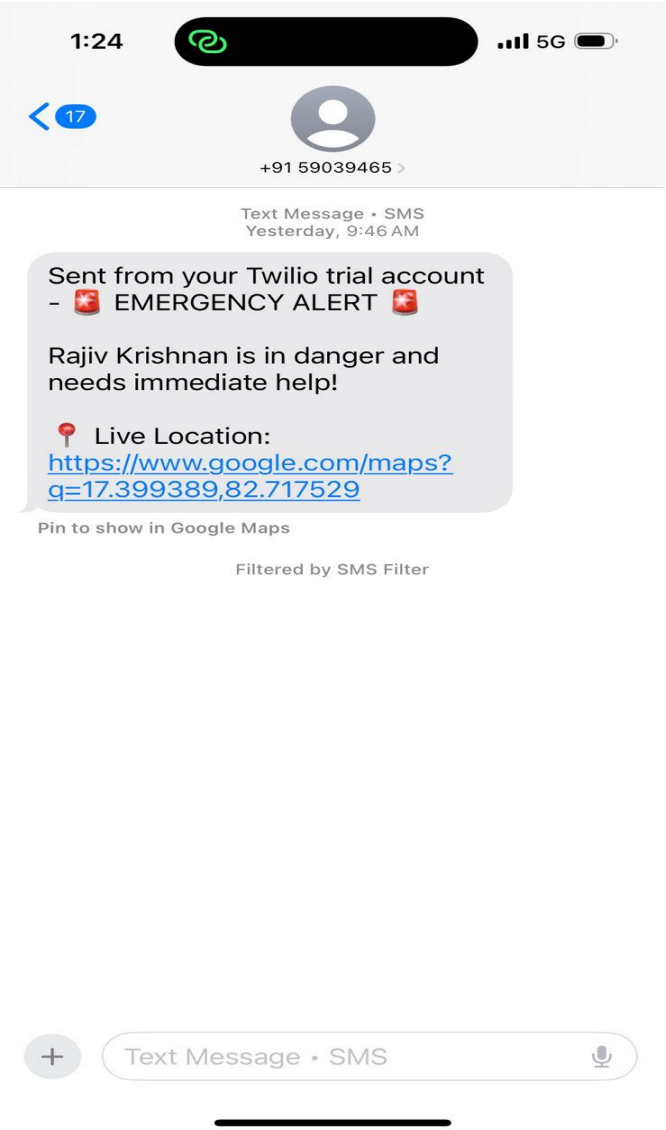


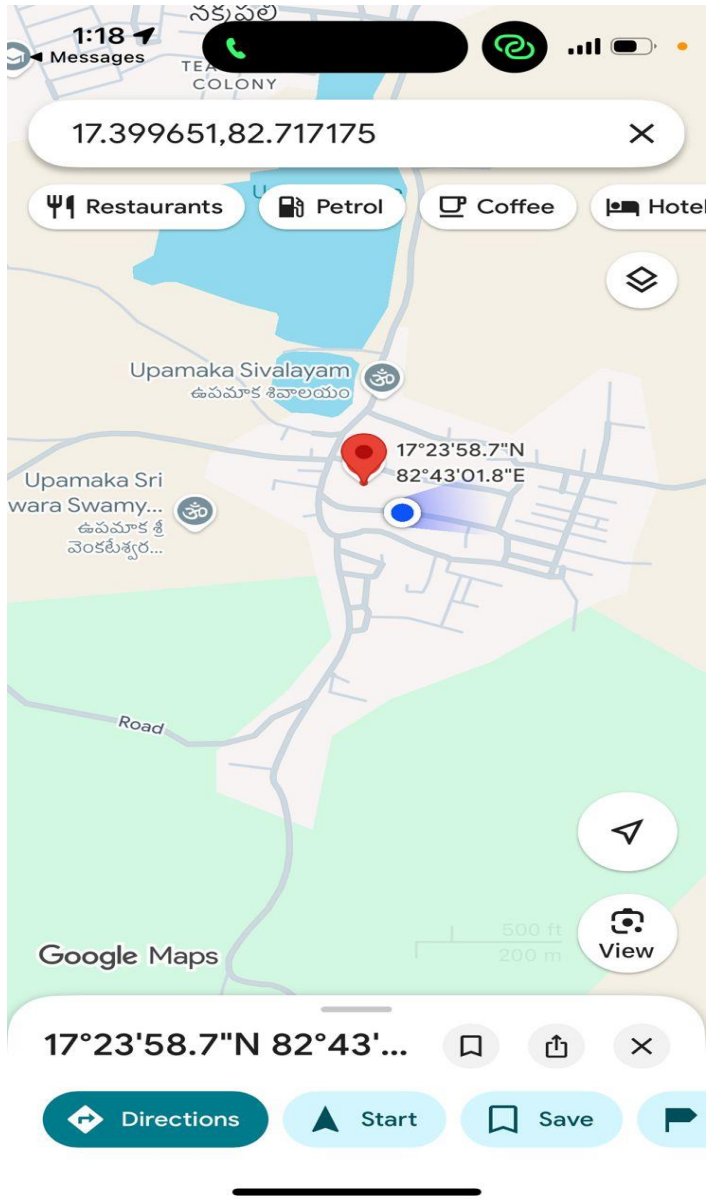

127.0.0.1:5000/login
Register
Login
Contact
← Back
Welcome Back!
Email
Enter your registered email
Password
Enter your password
Login
© 2025 Safety Bot


Logout
Contact
Emergency Sound Detection
Safety Bot is monitoring your surroundings for emergency sounds. If a distress sound is detected, alerts will be sent to your trusted contacts.
Start Listening
Stop Listening
Emergency Detected
© 2025 Safety Bot


127.0.0.1:5000/contact
← Back to Welcome
Contact Safety Bot
We are committed to your safety. For inquiries, feedback, or support, you can reach us via the following channels:
Email: safetybot@gmail.com
Phone: +91 82975 11784
Important Notice: Safety Bot is intended for emergency use only. Misuse of this system (false alarms, prank usage, or harassment) is strictly prohibited and may result in legal action under applicable laws. Please use responsibly.
© 2025 Safety Bot

```
✅ Model loaded
🎧 Background listener started
🔍 Probability: 0.107
🔍 Probability: 0.067
🔍 Probability: 0.066
🔍 Probability: 0.043
🔍 Probability: 0.097
🔍 Probability: 0.167
🔍 Probability: 0.447
🔍 Probability: 0.203
🔍 Probability: 0.231
🔍 Probability: 0.412
🔍 Probability: 0.588
🚨 EMERGENCY CONFIRMED
📤 SMS sent to +918555801462
📤 SMS sent to +919381966838
🔍 Probability: 0.228
🔍 Probability: 0.134
🔍 Probability: 0.080
🔍 Probability: 0.062
🔍 Probability: 0.096
🔍 Probability: 0.058
🔍 Probability: 0.055
🔍 Probability: 0.052
🔍 Probability: 0.051
🔍 Probability: 0.047
🔍 Probability: 0.100
🔍 Probability: 0.060
```





Impact Assessment

❑Technical Impact

- Introduces voice-driven emergency detection using speech recognition and audio analysis.
- Implements machine learning models to classify emergency severity in real-time.
- Enhances automation, accuracy, and reliability of emergency alert systems.

❑Social Impact

- Provides immediate help during critical situations, potentially saving lives.
- Reduces panic and dependency on manual assistance in emergencies.
- Improves safety for vulnerable individuals, including elderly and children.

❑Economic Impact

- Minimizes costs associated with delayed emergency response.
- Reduces unnecessary resource use through false alarm prevention.
- Supports scalable deployment for personal and institutional safety solutions.

Challenges Faced

- Collecting and labeling reliable emergency audio data while maintaining realism and safety.
- Differentiating between actual emergencies and normal loud sounds to reduce false alarms.
- Achieving accurate speech recognition under noise and panic conditions.
- Integrating real-time audio processing with Flask and SMS alert services.
- Ensuring system reliability, security, and misuse prevention during continuous operation.

Conclusion

This project demonstrates the effectiveness of an intelligent, voice-driven emergency detection and alerting system designed to provide timely assistance during critical situations. By integrating speech recognition, audio analysis, and machine learning techniques, the system reduces reliance on manual user actions and enhances emergency response capabilities. The inclusion of false alarm prevention mechanisms and a safe testing mode improves reliability and usability. Overall, the project highlights the potential of automated voice-based technologies to strengthen personal safety, improve emergency preparedness, and support faster, more dependable response in real-world scenarios.

Future Work

- Enable GPS-based location tracking to provide precise emergency location information.
- Integrate real-time language translation to support distress detection across multiple languages.
- Integrate automatic image capturing during emergency situations to provide visual context and secure transmission of captured images to registered mobile numbers or contacts.
- Combine image data with audio-based detection for improved emergency verification.

References

[1] Riza Kamelia; Hendra Kusuma. “Emergency Sound Classification and Visual Alert System for Enhanced Situational Awareness.” Proceedings of the 2024 International Conference on TVET Excellence & Development (ICTeD), IEEE, January 2025. Available: https://ieeexplore.ieee.org/abstract/document/10844649

[2] Hyoung-Gook Kim; Gee Yeun Kim. “Deep Neural Network-Based Indoor Emergency Awareness Using Contextual Information From Sound, Human Activity, and Indoor Position on Mobile Device.” IEEE Transactions on Consumer Electronics, August 2020. Available: https://ieeexplore.ieee.org/document/9163171

[3] Annmaria Mathew Shibu; Alphin Tom; Rohan Majo Mathew; Ryan Binny Mathews; Jacob Thomas; Renjith Thomas. “AI-Enhanced Emergency Call Centre Assistant and Situational Awareness for Optimized Emergency Response: Using Speech Recognition, NLP Mapping, NER.” IEEE, August 2025. Available: https://ieeexplore.ieee.org/abstract/document/11135720

[4] Ju-Hwan Lee; Hyoung-Gook Kim. “Self-Attentive Residual Temporal Convolutional Network-Based Indoor Emergency Awareness Using Contextual Information From Sound and Human Activity.” The Journal of the Acoustical Society of Korea, November 2024, pp. 637–643. Available: https://www.jask.or.kr/articles/xml/db8a/

Any Queries/ Questions?